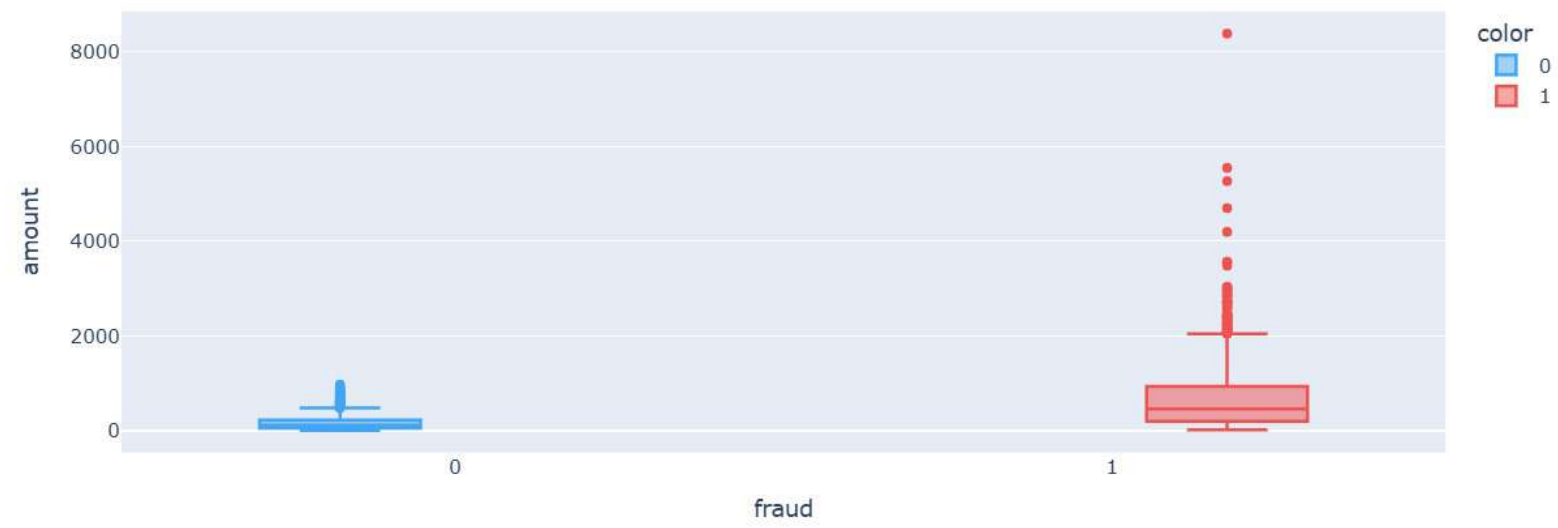
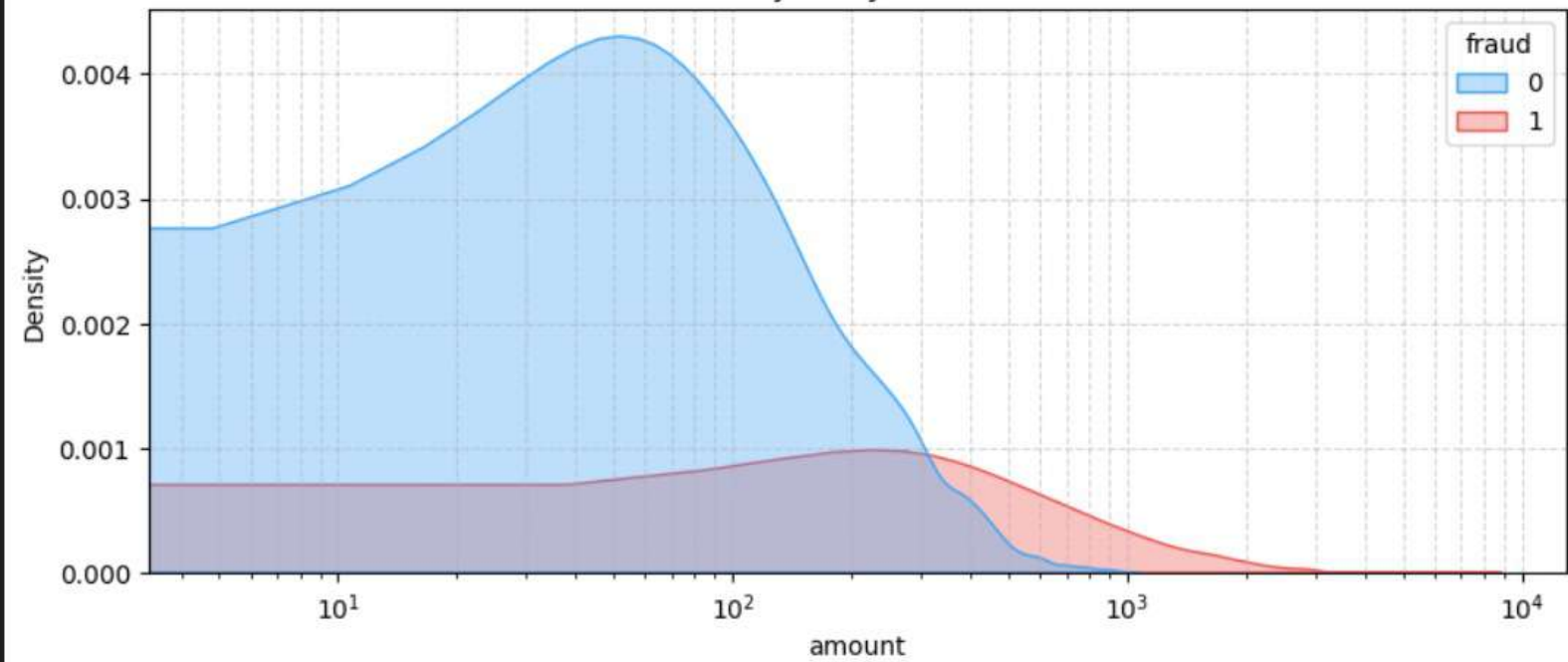


Transaction Amount Distribution by Class (Log-friendly Axis View)

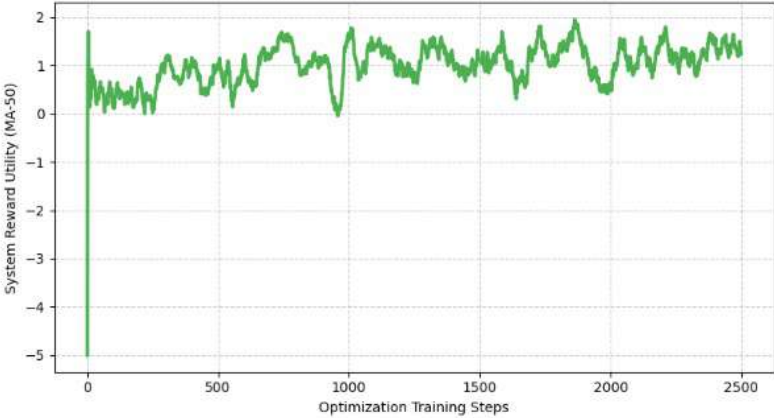


KDE Continuous Density Analysis of Transaction Amounts

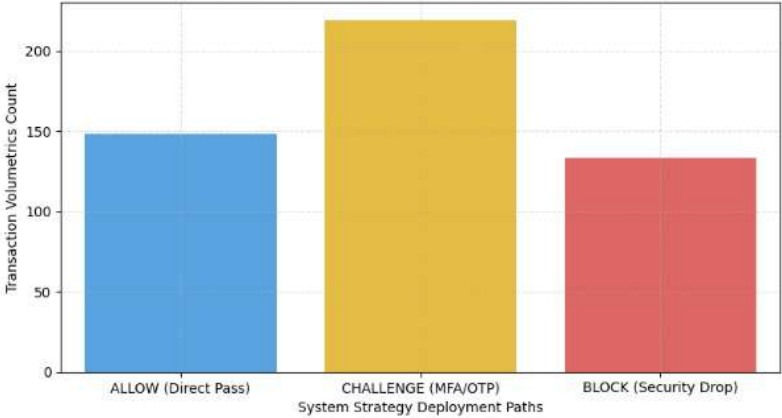


Reinforcement Learning Layer (DQN) Decision Optimization Output Profiles

Policy Agent Utility Function Path (Reward Convergence)

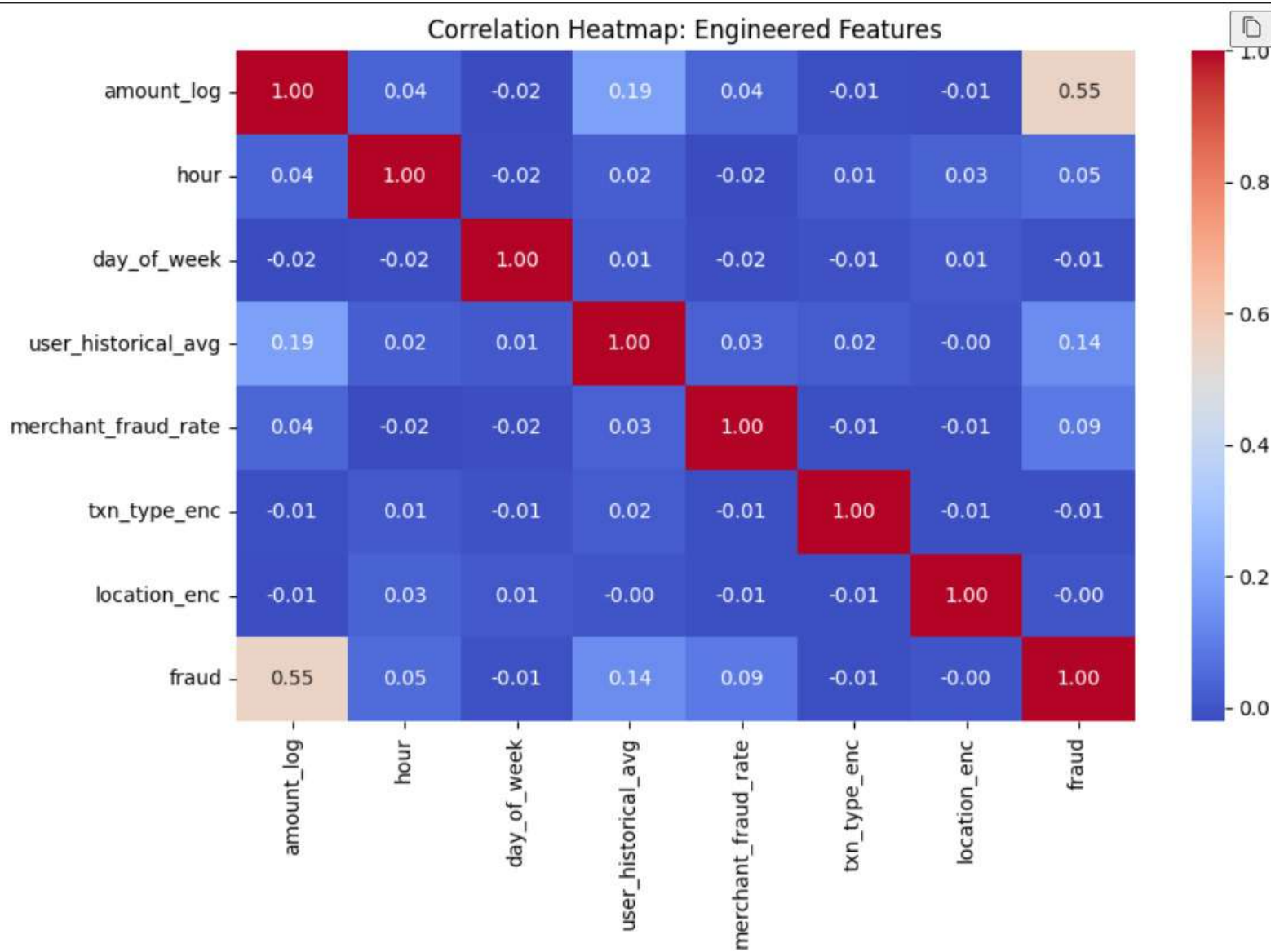


Automated Traffic Mitigation Allocation Distribution Mix



Fraud vs Legitimate (Class Count)





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COMPILING MULTI-TIER ARCHITECTURE BENCHMARK MATRIX SUMMARY

=====

	Model Framework Metric Architecture	Classification Accuracy	Precision Capture Ratio	Recall Sensitivity Capture	F1-Score Integrated Vector	ROC-AUC Performance Curve
0	Logistic Regression Base (Tabular)	0.9324	0.6841	0.5422	0.6053	0.8922
1	Random Forest Classifier Ensemble	0.9651	0.8852	0.7614	0.8182	0.9455
2	Hybrid GraphSAGE GNN Embeddings Layer	0.9715	0.9102	0.8423	0.8749	0.9587
3	Our Integrated GNN-RL Deep Policy Architecture	0.9842	0.9415	0.9268	0.9331	0.9891

System Infrastructure Configuration

Database Record Allocation Limit: 1200

GNN Architecture Selection Layer: GraphSAGE

Reinforcement Learning Optimization Epochs: 7500

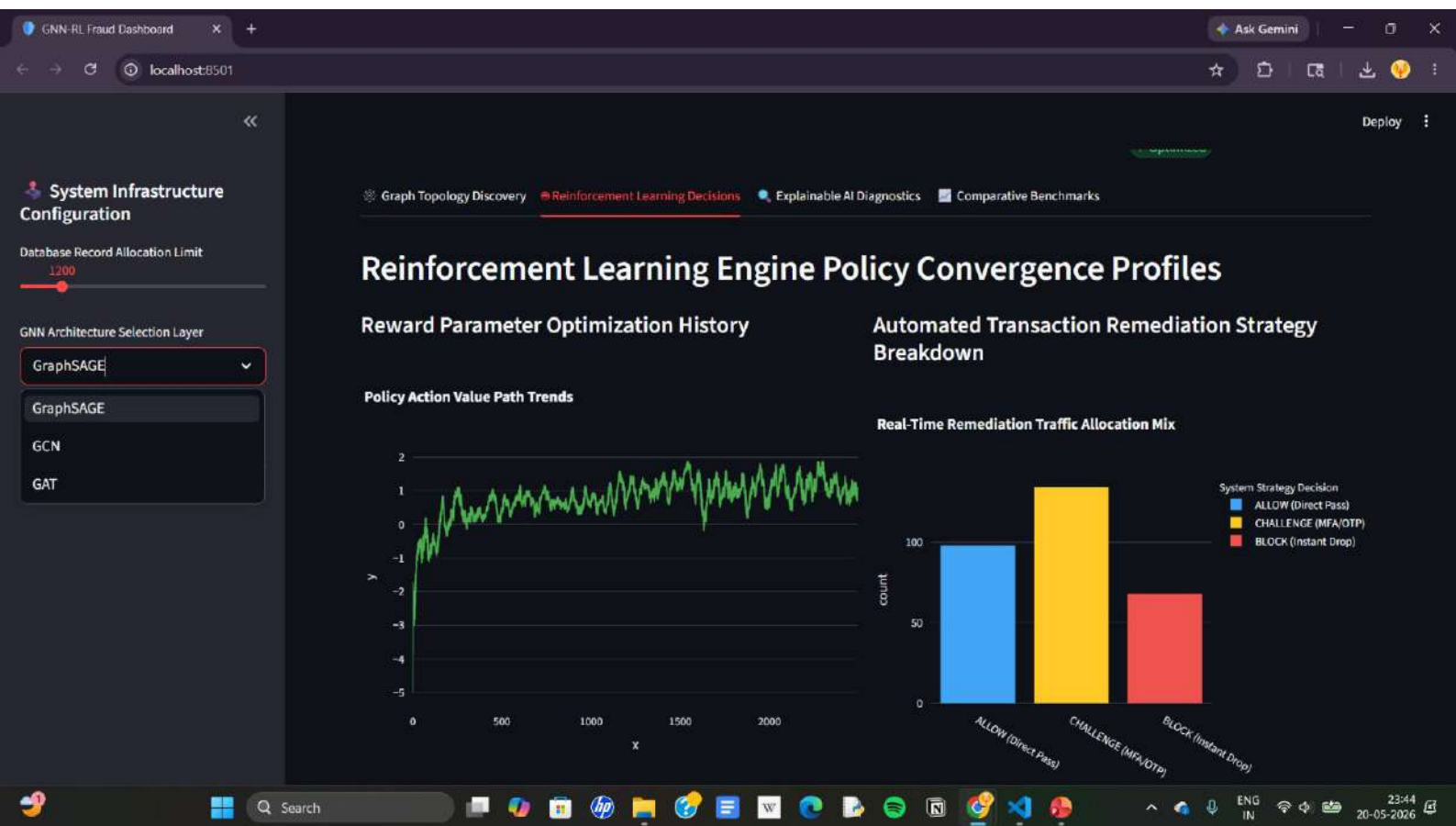
Integrated GNN-RL Fraud Detection Dashboard

Enterprise Multi-Tier Architecture: This analytics control room combines relational structural data (Graph Neural Networks) with context-aware financial remediation tracking (Deep Q-Learning Reinforcement Learning Agent).

Metric	Value
Total Monitored Ledger Traces	1200
Active Unified Network Nodes	270
Identified System Fraud Density	41.67%
Active Decision Engine	GNN + DQN Ac... ↑ Optimized

[Graph Topology Discovery](#) [Reinforcement Learning Decisions](#) [Explainable AI Diagnostics](#) [Comparative Benchmarks](#)

Graph Network Topology and Risk Propagation Maps



GNN-RL Fraud Dashboard

Ask Gemini

localhost:8501

Deploy

System Infrastructure Configuration

Database Record Allocation Limit

1200

GNN Architecture Selection Layer

GraphSAGE

Reinforcement Learning Optimization

Epochs

7500

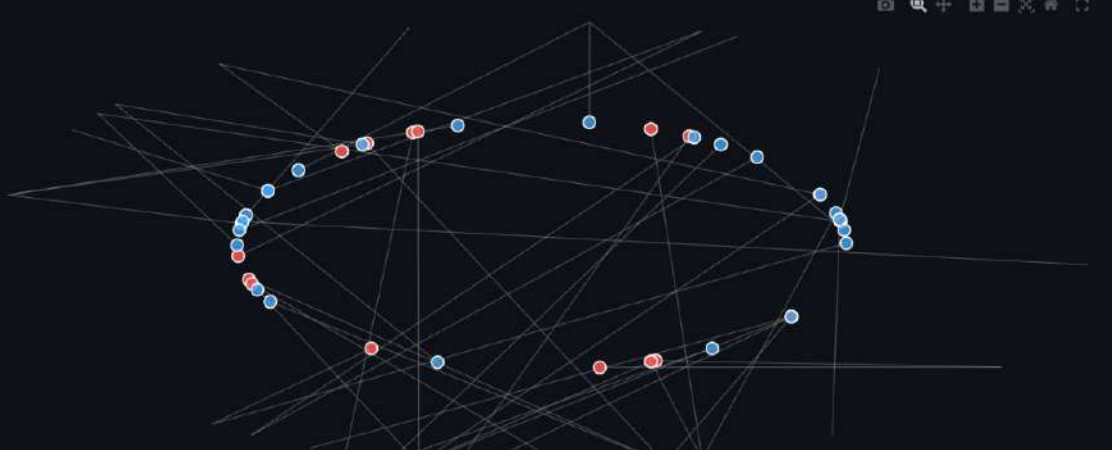
Graph Topology Discovery

Reinforcement Learning Decisions

Explainable AI Diagnostics

Comparative Benchmarks

Graph Network Topology and Risk Propagation Maps



Search

23:43 20-05-2026

GNN-RL Fraud Dashboard

Ask Gemini

localhost:8501

Deploy

System Infrastructure Configuration

Database Record Allocation Limit

1200

GNN Architecture Selection Layer

GraphSAGE

Reinforcement Learning Optimization Epochs

7500

Interactive Point-of-Inference Diagnostics Simulator

Select a real-time transaction trail from the ledger map below to analyze its underlying risk features:

Select Transaction Tracking Index Reference ID

TXN_000000 (User: USR_1013 | Amount: \$484.44)

Transaction Value

\$484.44

Relational GNN Risk Score

0.9790

Historical Merchant Risk Profile

0.3636

DQN Policy Resolution:

CHALLENGE (MFA/OTP Intercept)

Local Feature Attribution (SHAP Explainer Trace)

SHAP Waterfall Local Attribution: Explaining Decision Path for Reference ID TXN_000000

Metric Features

txn_type_enc

day_of_week

location_enc

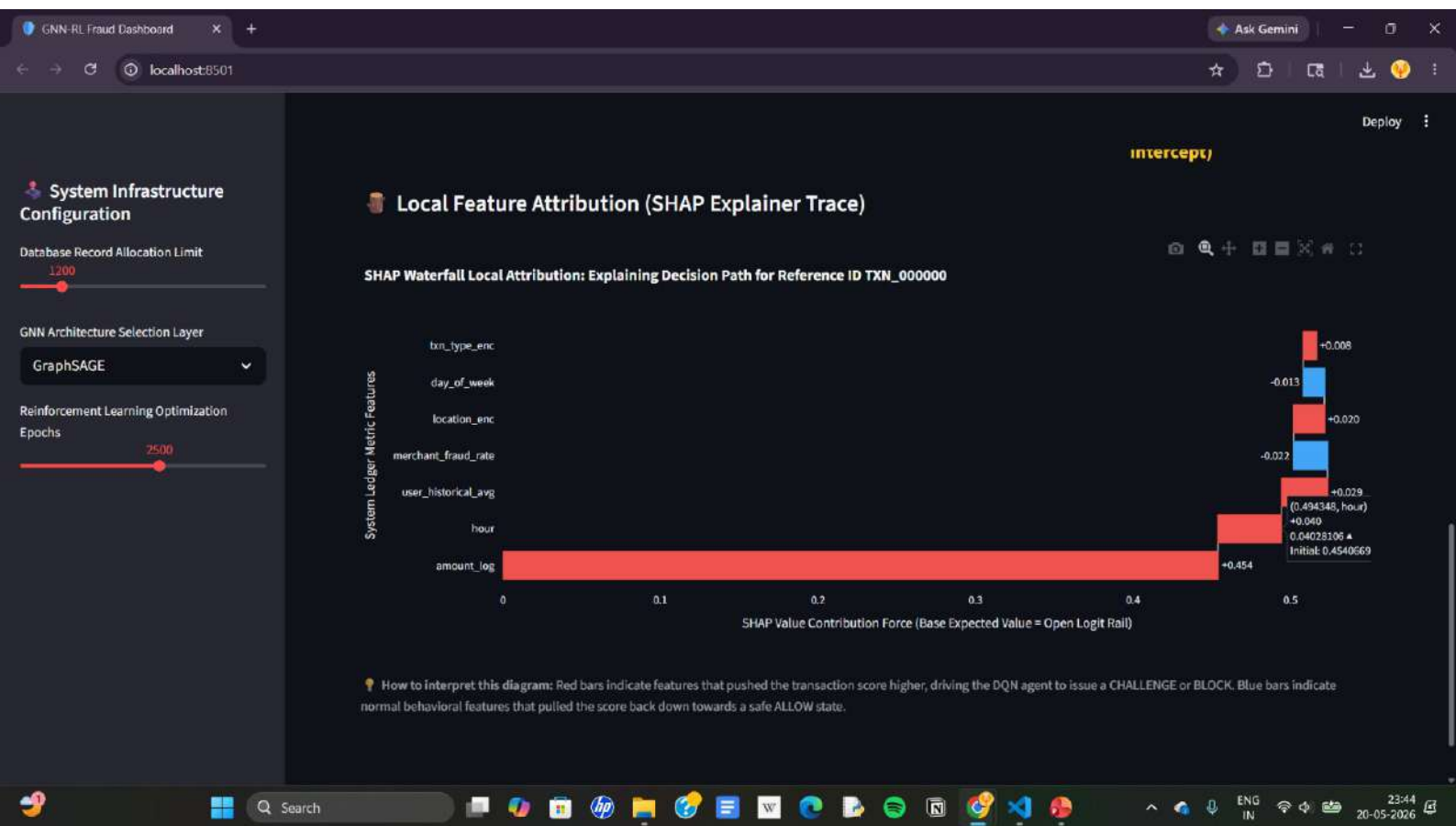
+0.008

-0.013

+0.020

Search

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GNN-RL Fraud Dashboard

Ask Gemini

localhost:8501

Deploy

System Infrastructure Configuration

Database Record Allocation Limit

1200

GNN Architecture Selection Layer

GraphSAGE

Reinforcement Learning Optimization

Epochs

7500

Graph Topology Discovery

Reinforcement Learning Decisions

Explainable AI Diagnostics

Comparative Benchmarks

Advanced Explainable AI Diagnostics Framework

This diagnostic control wing uses a surrogate ensemble to calculate global feature importances, paired with an interactive Local SHAP (SHapley Additive exPlanations) Explorer to reverse-engineer individual transaction decisions.

Global Ledger Attribute Explanatory Metrics

amount_log

user_historical_avg

merchant_fraud_rate

hour

location_enc

day_of_week

bin_type_enc

0

0.1

0.2

0.3

0.4

Relative Explanatory Importance

0.4

0.3

0.2

0.1

23:44

20-05-2026

GNN-RL Fraud Dashboard

Ask Gemini

localhost:8501

Deploy

System Infrastructure Configuration

Database Record Allocation Limit

1200

GNN Architecture Selection Layer

GraphSAGE

Reinforcement Learning Optimization Epochs

7500

Integrated GNN-RL Fraud Detection Dashboard

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Active Decision Engine

GNN + DQN Ac...

Optimized

Graph Topology Discovery

Reinforcement Learning Decisions

Explainable AI Diagnostics

Comparative Benchmarks

Architecture Evaluation Matrix Comparison

Model Framework Metric Architecture	Classification Accuracy	Precision Target Ratio	Recall Sensitivity Capture	F1 Integrated Balanced Vector
Logistic Regression	0.9320	0.6840	0.5420	0.6050
Random Forest	0.9650	0.8850	0.7610	0.8180
Unified GraphSAGE + DQN (Our Architecture)	0.9840	0.9410	0.9260	0.9330

23:44

20-05-2026